ISA5810 DM2026 Lab 1 — Master Notebook Questions

Fill in this file as you work through DM2026-Lab1-Master.ipynb. Submit this completed file converted to PDF together with the rest of your Lab 1 work.

Please do not change the style of this document.

Question numbers 1–10 follow the same order as the notebook (top to bottom).

Student Information

Please complete the fields below. For **English Name** and **Chinese Name**, you may fill in **only one** if you do not have the other.

- **English Name: Juan Francisco Verdugo Rivas**
- **Chinese Name: 維傑菲**
- **Email: s115065435@m115.nthu.tw.edu**
- **Student ID: 115065435**
- **University: National Tsing Hua University**

Instructions

These questions go with the Master Jupyter notebook (DM2026-Lab1-Master.ipynb). Each question is tied to a specific notebook section. Open the notebook, find **Question N (take home)**, and write your answer here.

We **strongly** suggest running the notebook to learn the lab, but the TAs will **not** audit whether you executed every cell to answer the questions in this document.

How to answer each question

1. Find the **notebook section** listed under the question (use the Table of Contents if needed).
2. Read the task; change parameters when asked (e.g. stop_words, thresholds, k, minSup).
3. Write a clear answer in the **Answer** space below each question.
4. Leave the question text unchanged; write only in the answer and disclosure fields.

Sources, honesty, and AI use

Acceptable vs. Unacceptable AI Usage: Generative AI should serve as an interactive tutor to guide your understanding, not as an automated solver completing the assignment for you.

- **Permitted (AI as Guide & Editor):**
  - Asking AI to explain unfamiliar theoretical concepts, discuss algorithmic trade-offs, or troubleshoot runtime errors in the notebook.
  - Polishing the grammar, phrasing, and readability of your own original, self-written drafts.

- **Prohibited (AI as Solver):**
  - Copying assignment questions directly into an AI tool to generate answers.
  - Using AI to fabricate experimental results or observations without executing the code in DM2026-Lab1-Master.ipynb.
  - Submitting machine-generated prose wholesale.

- **External material:** If you use websites, books, papers, or documentation beyond the notebook, cite them with **clickable links** (or DOI/ISBN when no URL exists).

- **When AI was used:** Your answer is acceptable if you (1) **disclose** AI use and (2) include **at least one external link** that validates the information in your answer (official docs, textbook, paper, or reputable tutorial — not the AI chat alone as the only source).

- **When AI was not used:** You may write "No" under **AI used** or leave it blank only if you truly did not use AI for that question. References are still encouraged if you looked anything up.

Under each question, complete:

- **AI used for this answer:** Yes / No
- **If yes, how:** (e.g. brainstorming, grammar, concept explanation)
- **External references (links):** *(required when AI = Yes; encouraged otherwise)*

Write in English. Keep answers concise and specific.

Questions

Question 1: Data Filtering & Domain Relevance

Notebook: Section 3.2 — Question 1 (take home)

Task: Filter the dataframe **X** for the category **sci.med** and check its length.

Questions:

1. How many documents are in this specific category?
2. From a data mining perspective, why is isolating specific categories a crucial first step before applying algorithms like frequent pattern mining?

Answer: There are 594 documents in the sci.med category. In the case of frequent pattern mining, it can help to understand the frequency by which element appears in the dataset. It will later help with operations like support counting to determine the frequency of occurrence of some itemsets. On the other hand, isolating specific categories helps to work with a single category of elements (which may be necessary depending on the objective of the study).

AI used for this answer: No

If yes, how:

External references (links): P.-N. Tan, M. Steinbach, and V. Kumar, *Introduction to Data Mining*. Pearson, 2020.

Question 2: Handling Missing Data

Notebook: Section 4.1 — Question 2 (take home) (after the **dropna()** demonstration)

Task: In Section 4.1, a dummy record with a missing value is injected and then removed using the **dropna()** function.

Question: If 40% of documents were missing the **category_name** label, why would simply dropping these rows (with **dropna()**) be a flawed data mining strategy? Name one objective alternative so the text data can still be utilized.

Answer: Considering that 40% of the data is an important section of the whole dataset, it is not convenient because of two reasons. First, the data describes the patterns, attributes, and general behavior of a measured phenomenon. Erasing a big portion of the data could eliminate important data points that describe real behavior. Therefore, conclusions inferred from data analysis could have a bias. Second, missing values could be missing for other reasons. For example, in a survey there are missing values of names, a profound analysis could indicate that certain portions of the sample didn't want to give their names, maybe because those people belong to vulnerable populations who are in danger. So first, data that is missing must be understood why is missing, and after understanding it a decision can be made on how to process it. This is because, even missing data could be relevant for some analysis. An alternative is to estimate missing values, that is less complicated with ratio data. Doing a regression could help to calculate or estimate missing values according to other attributes.

AI used for this answer: No

If yes, how:

External references (links):

Question 3: Data Sampling Risks

Notebook: Section 5.1 — Question 3 (take home)

Task: In Section 5.1, the code uses **X_sample = X.sample(n=1000)** to create a subset. Compare the bar chart of the original category distribution to the sampled distribution.

Questions:

1. Do the exact category proportions perfectly match the original?
2. What is the specific risk of simple random sampling on a highly imbalanced dataset?

Answer: It is nearly the same category proportions from the original dataset compared to the sample. The most frequent category in the original dataset is comp.graphics. In this sample that category was slightly inflated (in the original dataset it was 26.54% and in the sample was 27.80%), and the rest of the categories have a slightly lower percentage. In this case the inflation is small, but it represents the risk of simple random sampling. In a more imbalanced dataset overrepresentation of already large categories can happen in the sample and categories with few occurrences may disappear or be underrepresented.

AI used for this answer: Yes

If yes, how: Grammar check

External references (links):

Question 4: Text Vectorization and Matrix Sparsity

Notebook: Section 5.3 — Question 4 (take home)

Task: In Section 5.3, the **CountVectorizer** converts text into a document-term matrix using **stop_words='english'**. Change this parameter to **stop_words=None** and re-run the cell to check the shape of **X_counts**.

Questions:

1. Does the total number of features (columns) increase or decrease?
2. Since stopwords are highly common words that appear in almost all documents, what effect does adding them back have on the overall sparsity (the percentage of zeros) of the matrix?

Answer: The total number of features increases to a total of 306. If stopwords are kept in the dataset, the sparsity of the matrix would be lower because these words will probably appear constantly in all the documents of the dataset. Therefore, having more words with

repetitions between documents means the 0s in the sparsity matrix would share a less percentage of the total.

AI used for this answer: No

If yes, how:

External references (links):

Question 5: Feature Selection — Variance Thresholding

Notebook: Section 5.3.1 — Question 5 (take home)

Task: In the variance filtering section, re-run **variance_filter** using threshold values **threshold = 0.0, threshold = 0.01, and threshold = 0.05**. Record the number of features marked as **keep == True**.

Questions:

1. What happens to the total size of the vocabulary as the variance threshold increases from 0.0 to 0.05?
2. In terms of document frequency, what kinds of words have variance close to zero and are eliminated first?
3. Why is eliminating near-zero variance features considered a safe way to reduce dimensionality without harming model accuracy?

Answer:

Threshold	Number of terms kept (keep == True)
0.0	35,482
0.01	5,089
0.05	1491

The number of words kept from the vocabulary decreases as the variance threshold increases. The words that have a variance close to zero are the ones that have a similar frequency of appearances in all the documents of the dataset. This could mean that they rarely appear, so the frequency normally is 0, or they are present in all the documents with nearly same number of appearances in each one. Eliminating near-zero values is a safe way to reduce dimensionality because it does not involve transforming data from its original state. First, eliminating words is only a form of prioritizing the data that is useful for the analysis, some words are kept and other ones are dropped. Therefore, all the data kept in the model hasn't suffered any transformation, it was only prioritized and it is still nearly raw and reflects truthfully the reality of what is measured.

AI used for this answer: No

If yes, how:

External references (links): P.-N. Tan, M. Steinbach, and V. Kumar, *Introduction to Data Mining*. Pearson, 2020.

Question 6: Feature Correlation — Linear (Pearson) vs. Monotonic (Spearman)

Notebook: Section 5.3.2 — Question 6 (take home)

Task: Run the correlation filter against the binary indicator for the **comp.graphics** class. Inspect the top 10 features sorted by **pearson_r** and compare them to the top 10 features sorted by **spearman_r**.

Questions:

1. Are the top-correlated words generic stop words or domain-specific terminology for graphics?
2. What is the mathematical difference in what **Pearson** correlation measures versus what **Spearman** rank correlation measures?
3. In what specific scenario would a word have a high **Spearman** rank correlation with a topic, but a significantly lower **Pearson** correlation?

Answer: In the case of the Pearson correlation, the top word is "thanks" which is a stop word. In the case of the Spearman correlation the first word is "graphics" which is a domain-specific terminology. The Pearson correlation measures the strength of a linear association between two variables by summing the difference of an attribute with its mean divided by the product of the squared differences from the object means. Therefore, Person heavily relies on the mean for the calculation. The Spearman's correlation needs first each attribute to be ranked and then the difference of the rankings between two attributes are elevated to the square and added. Later, it is divided by the total number of cases multiplied by the total number of cases elevated to the square minus 1. Spearman's correlation doesn't depend on how linear the relationship between the two variables is. For two variables to correlate with Pearson correlation, they need to move with a similar variance with the mean, meanwhile with Spearman's correlation they only must coincide with the direction of how they move.

A word could have a high Spearman's correlation and lower Pearson's correlation in the next scenario: There are two words: w1 and w2 that are ranked. If w1 and w2 both increase their values, the correlation would be high. However, this increase doesn't need to be the same between the two words. Meanwhile, for Pearson's correlation to be high, both words would need to increase at a similar rate. So, in this scenario, w1 could increase its ranking every time w2 appears, but w2 doesn't need to increase with the same rate. That would return a high Spearman's correlation and a lower Pearson's correlation.

AI used for this answer: Yes

If yes, how: To answer this question I had to understand what the one-vs-rest transformation for each of the four classes was. It was hard to understand it so I asked the AI to explain to me how that works and how that computation worked in the code.

External references (links):

https://statistics.laerd.com/statistical-guides/pearson-correlation-coefficient-statistical-guide.php

https://statistics.laerd.com/statistical-guides/spearmans-rank-order-correlation-statistical-guide.php

https://www.sciencedirect.com/topics/computer-science/pearson-correlation

---

Question 7: Frequent Pattern Mining — Algorithm & Parameter Dynamics

Notebook: Section 5.4.2 — Question 7 (take home)

Task: In the Frequent Pattern Mining section, run **Top-K FAE** with **k = 100 and k = 500**. Then, run **MaxFPGrowth** using minimum support thresholds **minSup=9 and minSup= 3**.

Questions:

1. How does decreasing **minSup** (from 9 to 3) in **MaxFPGrowth** affect the execution runtime and the total number of discovered patterns?
2. How does the parameter k in **Top-K FAE** differ fundamentally from **minSup** in **MaxFPGrowth** when deciding which patterns to keep?
3. Why are "maximal" frequent patterns structurally more concise than standard frequent patterns?

Answer (1) — Runtime and pattern count when lowering minSup:

Running minSup with 3 runs in 0.867. Running it with 9 runs in 0.049. So incremeanting the filtering of minSup the execution time lowers because there are few patterns to mine.

Answer (2) — Top-K k vs MaxFPGrowth minSup:

The difference consists in the number of final patterns after the operation. In the Top-K FAE the user determines the number of patterns the final output will have, so it is more controlled. On the other hand, with minSup the user determines the minimum support threshold. Therefore, the final number of patterns is not controlled by the user and is the result of the patterns that comply with the minSup value.

Answer (3) — Maximal vs standard frequent patterns:

The maximal approach selects the most frequent patterns and the largest. Therefore, a pattern will be selected only if it reaches a maximal length without reducing the frequency. In a standard frequent pattern, the length does not matter, so any small pattern that has a high frequency will be selected. The problem is that small patterns have a higher probability of having high frequencies because they are simpler, so few words in a pattern increase the chances of it appearing. This would make the measure sensible to noise made by many small patterns appearing. The maximal approach avoids that risk by selecting more complex and largest patterns which in theory should be more difficult to appear with high frequency in a random manner. So, a large pattern with high frequency could be a relevant source of information.

This analysis with the hypothesis that smaller and simpler patterns have a higher probability of appearing and could be noise was made without an external source. It was induced by the graphs from section 5.4.2 of the notebook.

AI used for this answer: No

If yes, how:

External references (links):

Question 8: Dimensionality Reduction — Motivation

Notebook: Section 5.5 — Question 8 (take home) (before PCA/t-SNE code)

Task: After text vectorization, the **X_counts** matrix creates tens of thousands of distinct features (words).

Questions:

1. If we want to visually map how these documents group by category on a scatter plot, why is it mathematically and visually impossible to plot the raw **X_counts** matrix directly?
2. What specific category of data mining algorithms (referenced in the helper files) must be applied first?

Answer: It is impossible to map X_counts as there are high number of dimensions in the dataset. For a scattered plot to work, it would require only a 2-dimension or 3-dimension dataset. Mathematically is hard to make graphs of four or more dimensions. The category of data mining algorithms to be applied are the ones made for dimension reduction.

AI used for this answer: No

If yes, how:

External references (links):

Question 9: Dimensionality Reduction — Linear (PCA) vs. Non-linear Manifolds (t-SNE/UMAP)

Notebook: Section 5.5 — Question 9 (take home) (after 2D/3D plots)

Task: In the 2D/3D Dimensionality Reduction section, compare the scatter plot projection generated by **PCA** against that generated by **t-SNE** or **UMAP**.

Questions:

1. Which method **(PCA or t-SNE/UMAP**) shows distinct, separated cluster islands for the 4 document categories, and which shows heavy overlap around the center?
2. Why does **PCA** struggle to separate high-dimensional text categories compared to **t-SNE** or **UMAP**?
3. Why must data analysts be careful not to over-interpret the physical distance between separate clusters in a **t-SNE** plot?

Answer: t/SNE/UMAP create the most distinct separated cluster islands, meanwhile PCA heavily overlaps the data in the center. PCA struggles to separate high-dimensional text categories because it does not search to preserve local structure but it searches for the directions that explain more of the variance. For that reason, in the 3D plots the points

follow invisible lines. On the other hand, t-SNE and UMAP search to preserve the structure by accounting vicinity with other points. Data analyst should be careful about interpreting the distance of t-SNE because it is an algorithm destined to preserve local structure. As Scikit-Learn's guide (link in external sources) explains, this algorithm preserves the structure of points that are neighbors but ignores in some way the global structure. So, distance between cluster's is not exactly calculated and are not as reliable as the distance between local neighbors.

AI used for this answer: Yes

If yes, how: I was struggling trying to understand how PCA is calculated. So I asked ChatGPT to explain it to me step by step. I did that after reading the following external references.

External references (links):

https://scikit-learn.org/stable/modules/manifold.html#t-sne

https://www.geeksforgeeks.org/machine-learning/implementing-pca-in-python-with-scikit-learn/

https://www.geeksforgeeks.org/data-analysis/principal-component-analysis-pca/+

https://www.geeksforgeeks.org/machine-learning/umap-uniform-manifold-approximation-and-projection/

https://www.geeksforgeeks.org/machine-learning/ml-t-distributed-stochastic-neighbor-embedding-t-sne-algorithm/

Question 10: One-Hot Binarization

Notebook: Section 5.6 — Question 10 (take home)

Task: In Section 3.1, examine **X['category']** (integers 0, 1, 2, 3). Then look at the output of **LabelBinarizer** in Section 5.6, where each category is converted into a binary vector (e.g., [1, 0, 0, 0]).

Questions:

1. How many binary columns are created to represent the 4 categories?
2. If an algorithm (like linear regression or KNN) is given the raw integer column **(0, 1, 2, 3)** as input, what false mathematical assumption does it make about the categories?
3. How does one-hot binarization eliminate this false assumption?

Answer: It creates four columns, one for each category. The problem of having a raw integer column from 0 to 3 is that it could assume the variable is a continuous one where values increase with a real distance. For example, the distance between 0 and 3 is greater, so an algorithm could consider that distance as real. This data is nominal, where distance does not matter. Binarization allows avoiding that because it creates four separated columns with 0s and 1s, so there is not a increasing distance between numbers that could confuse an algorithm.

AI used for this answer: No

If yes, how:

External references (links):

https://www.geeksforgeeks.org/machine-learning/assumptions-of-linear-regression/

https://www.geeksforgeeks.org/machine-learning/k-nearest-neighbours/
